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Understanding the process-microstructure correlations for tailoring the mechanical properties of L-PBF produced austenitic advanced high strength steel

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ABSTRACT

In this work, the additive manufacturing technique of laser powder bed fusion (L-PBF) was used to build up X30Mn21 austenitic advanced high strength steel (AHSS) samples. Different L-PBF process parameters were used to understand the correlation between process, microstructure, texture, and mechanical properties. The influence of build platform preheating (200 °C–800 °C), laser speed (550 mm/s - 950 mm/s) and scan strategy (bi-directional continuous and Mark&Sleep (M&S)) on grain size, grain morphology, size of solidification cells, dislocation density, and texture was studied. Local solidification parameters in the melt pool e.g. cooling rates, temperature gradients and solidification velocities were simulated by a FEM heat flow model and correlated with the solidification microstructure. By using SEM/EBSD analysis and tensile testing, the mechanical properties of the AHSS were assessed by considering microstructural aspects. It was found that AHSS, produced with higher laser speeds and an alternative M&S scan strategy, revealed a reduced grain size and texture intensity. This was attributed to a partial columnar to equiaxed transition (CET), as well as a significantly increased density of geometrically necessary dislocations. Preheating of the build platform promoted columnar grain growth with a more pronounced texture, low dislocation densities, and reduced yield strength. The influence of cooling rate, temperature gradient and solidification velocity on microstructural and textural evolution is discussed based on fundamental solidification theories.

1. Introduction

Laser powder bed fusion (L-PBF) reveals the highest maturity and industrial application of all metal additive manufacturing (AM) technologies [1–3]. L-PBF has successfully been used to produce AISI 316 L [4], 17-4 PH [5], maraging steels [6] and new promising advanced high strength steel (AHSS) classes, such as high manganese steels (HMnS). HMnS offer the additional deformation mechanism of transformation-induced plasticity (TRIP) and twinning-induced plasticity (TWIP) [7]. Lightweight lattice structures with high specific energy absorption potential have been produced in previous studies [8] by combining the geometrical freedom of L-PBF to produce periodic lattice structures with interesting weight specific properties [9] and the unique mechanical properties of HMnS.

HMnS are highly alloyed austenitic steels with typically 15–30 wt% manganese, up to 1 wt% carbon, and up to 3 wt% aluminum. The low stacking fault energy (SFE) of HMnS determines the dominant secondary deformation mechanisms, e.g. TRIP and/or TWIP next to dislocation glide. TRIP is observed until an SFE of ca. 15 mJ/m^2 and TWIP between 15 mJ/m^2 to 45 mJ/m^2 [10]. Due to the additional deformation mechanisms, HMnS reveal very high strain hardening rates, which can be adjusted by changing the alloy composition [11]. For instance, carbon contributes to the solid solution hardening effect ($\sim$ 250 MPa/wt%), enhances the austenite stability and increases the SFE. Manganese is primarily used to stabilize the austenite phase at room temperature. Alloying with manganese can decrease the SFE ($<$ 16 at %) or increase the SFE ($>$ 16 at %). Addition of aluminum reduces the alloy's density, inhibits dynamic strain aging in C-alloyed HMnS and increases the SFE by $\sim$ 8 mJ/m^2 per wt% Al [12–14].

In L-PBF, the fast cooling rates [15], substantial temperature

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gradients [16] and strong melt pool dynamics [17,18] facilitate the evolution of fine, as-built microstructures [19]. Due to these unique process-related peculiarities, L-PBF manufactured HMnS steels show outstanding mechanical properties of high yield strength, tensile strength and ductility already in the as-built condition [8]. This combination of mechanical properties is difficult to achieve with conventional production routes, like ingot or continuous casting, without post treatments [13,20]. Due to the high cooling rates during L-PBF, segregation effects of the alloying elements can be reduced [20,21] which can decrease necessary post-processing steps to homogenize the microstructure [22].

Thus far, several studies have been performed that intended to reveal the correlation between AM process parameters and microstructure evolution. Niendorf et al. [23] produced a highly anisotropic microstructure with strong texture by variation of the laser power. Furthermore, Martin et al. [24] reported the evolution of an isotropic microstructure with fine equiaxed grains by introducing nanoparticles that serve as heterogeneous nucleation sites. Roehling et al. [25] modulated the laser beam profile ellipticity to influence grain morphology. Nevertheless, in most of these studies, the process parameters are varied within a small range, which prohibits the establishment of a conclusive picture of the process-microstructure correlations. The parameter window for L-PBF is typically kept small in order to ensure geometrical accuracy, reduce the probability for defect formation e.g. keyhole formation, balling or delamination [26] or due to limitations in the adjustability of the process parameters in industrial L-PBF machines.

The aim of the present study is to exploit a wide L-PBF process parameter window to investigate the correlation between process parameters, microstructure, crystallographic texture and mechanical properties of additively manufactured HMnS. The L-PBF process parameters, build platform preheating temperature, laser speed and the off-time of the laser between two adjacent scan tracks were varied to influence heat input and accordingly the solidification parameters, i.e. cooling rate, thermal gradient, and solidification velocity. To quantify the solidification parameters, finite element method (FEM) simulations of the L-PBF process of HMnS were performed. In order to investigate the influence of the solidification parameters on the microstructure, scanning electron microscopy (SEM) in combination with electron backscatter diffraction (EBSD) was used, and tensile tests were carried out to analyze the mechanical properties. Influential factors of microstructure formation such as cooling rate, temperature gradient, and solidification velocity were discussed based on fundamental solidification theories and correlated with mechanical properties. As of yet, a detailed analysis of the influence of different L-PBF process parameters on the microstructure and mechanical properties of AHSS in combination with subsequent correlation of simulated solidification parameters has not been reported.

2. Material and methods

2.1. Material and processing

The X30Mn21 HMnS powder (provided by thyssenkrupp Raw Materials GmbH, Germany) was ingot-casted and argon gas-atomized by the EIGA-technique (Electrode Induction Melting Gas Atomization). The chemical composition of the metal powder was measured by inductively coupled plasma optical emission spectrometry (ICP-OES) and the combustion method (Table 1). The metal powder particles revealed a spheroidal shape (Fig. 1a) and were sieved, and air separated to achieve a suitable particle size distribution. The number distribution of the particle size (Fig. 1b) was measured by optical image analysis of 402,409 particles according to ISO 13322-1 using a Morphology G3 particle analyzer (Malvern Panalytical, UK). The average particle size was 14.94 μm, and the first decile (D10) was 9.95 μm, the median value (D50) was 12.99 μm, and the last decile (D90) was 22.84 μm. An apparent powder density of 4.11 g/cm^3 was measured using the hall funnel method according to ISO 3923-1. An avalanche angle of 35.53 ° was measured using a revolution powder analyzer (PS Prozesstechnik GmbH, Germany). During the L-PBF process, the chemical composition was changed due to Mn vaporization (Table 1). The stacking fault energy (SFE) of the X30Mn21 HMnS in the v750 condition was calculated to be ca. 10 mJ/m^2 using a thermodynamic model from Saeed-Akbari et al. [10]. A detailed analysis of the deformation mechanisms is reported in [20].

Table 1
Chemical compositions of X30Mn21 metal powder and L-PBF produced bulk material (condition v750, see Table 2). Fe, Mn, Al, Cr, Ni were measured by inductively coupled plasma optical emission spectrometry (ICP-OES), whereas C and O were determined by combustion method. All contents are given in wt %.

State	Fe	Mn	C	Al	Cr	Ni	O
metal powder	base	21.0	0.33	0.03	0.15	0.08	0.10
as-built (v750)	base	18.7	0.31	0.03	0.17	0.10	0.04

HMnS bulk specimens for mechanical and microstructural analysis were produced using an Aconity Mini L-PBF machine (Aconity-3D GmbH, Herzogenrath, Germany) equipped with a Yb:YAG-laser (400 W). The build platform was cylindrical with a radius of 140 mm and a height of 190 mm and could be heated to 800 °C. A laser power of 120 W, a gaussian laser intensity profile with laser focus diameter of 80 μm, a hatch spacing of 70 μm and a layer thickness of 30 μm were used in combination with a bidirectional scan strategy. A rotation of 33 degrees between each layer was chosen. The build chamber was flooded with argon gas (purity ≥ 99.996%) with a flow rate of 3 L/min to achieve a build chamber excess pressure of 100 mbar and an average oxygen concentration of ca. 100 ppm during the L-PBF process. The L-PBF process parameters of laser speed, build platform preheating temperature, and off-time of the laser between two laser hatch scans (Mark&Sleep, M&S) were varied according to Table 2. All process parameters were suitable for the production of dense bulk specimens with a relative density of ≥ 99.9% (Table A3, Fig. A1), as obtained by optical measurement of the residual porosity. Only virgin metal powder was used.

2.2. Specimens and experimental characterization techniques

Microstructural analysis was performed parallel to the building direction (BD) in the middle of the cross-section of cubic samples with an edge length of 10 mm. Cubic samples were prepared for microstructural analysis with a GeminiSEM field emission gun (Carl Zeiss AG, Germany) by mechanical grinding (up to 1200 SiC grit paper), mechanical polishing (3 and 1 μm diamond suspension) and etching with a Nital (5%) solution. The line intercept method, after DIN EN ISO 643 [27], was used for analysis of the solidification cell size. The

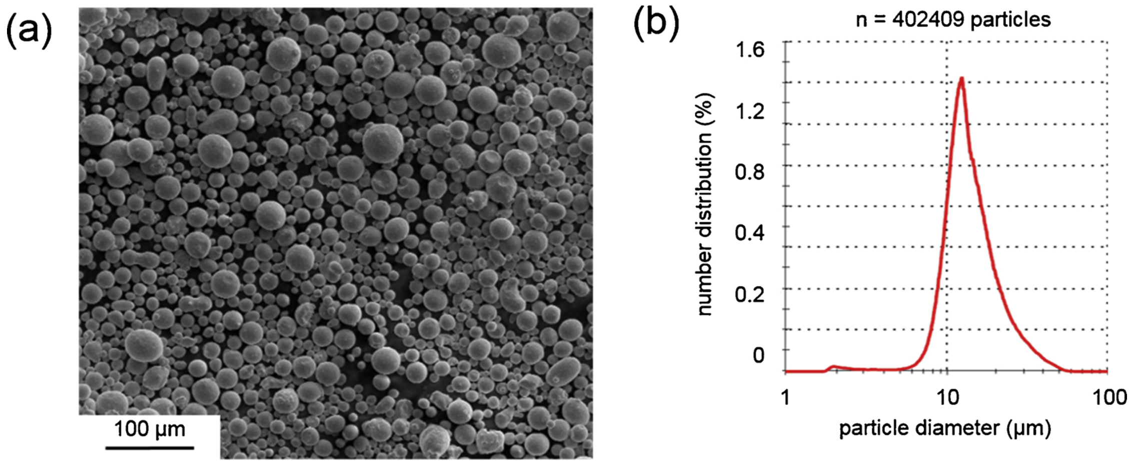

Fig. 1. (a) SEM micrograph of the X30Mn21 metal powder. (b) Corresponding number distribution of particle size measured by optical image analysis.

Table 2
L-PBF process parameters for the production of bulk specimens using X30Mn21 metal powder.

L-PBF process parameter	laser speed (mm/s)	preheating temperature of the build platform (°C)	laser off-time between two adjacent laser tracks (ms)
v550	550	–	0
v750	750	–	0
v950	950	–	0
200 °C	750	200	0
400 °C	750	400	0
600 °C	750	600	0
800 °C	750	800	0
M&S	750	–	50

mechanically polished samples were further electropolished with LectroPol-5 electrolytic polishing machine (Struers GmbH, Germany) with an A2 electrolyte (Struers GmbH, Germany) for EBSD analysis. Electropolishing was conducted for 20 s at 28 V at room temperature. For EBSD measurements, the same SEM was used with a 60 µm aperture, an accelerating voltage of 20 kV, and a NordlysNano (Oxford Instruments plc, Great Britain) detector with a step size of 500 nm. The EBSD measurements were used to examine the existing phases, the average grain size, the average grain aspect ratio defined as the ratio of length/width of the grains, and the grain orientation in relation to the BD. The EBSD data was processed and analyzed with the MATLAB toolbox MTEX [28]. The grain size was defined as the diameter of a circle with an area equivalent to the measured grain. Smoothing of EBSD data was performed with the MTEX half-quadratic filter, which preserves inner grain structures. High-angle grain boundaries were defined by a misorientation angle $\theta \geq 10^{\circ}$ between two measured points. Geometrically necessary dislocation (GND) densities were calculated from EBSD measurements with the MATLAB toolbox MTEX following the approach by Pantleon [29]. For determination of mechanical properties at room temperature an uniaxial tensile testing machine of type Z100 (Zwick/Roell GmbH & Co. KG, Germany) was used at a constant strain rate of 10^{-3} s^{1}. During tensile testing, the strain was measured with a videoXtens extensometer (Zwick/Roell GmbH & Co. KG, Germany) and the force with an Xforce load cell (Zwick/Roell GmbH & Co. KG, Germany). The B6 × 30 tensile samples, with a gauge diameter of 6 mm and a gauge length of 30 mm, were mechanically turned from L-PBF produced cylindrical bars with a diameter of 12 mm and a length of 82 mm. Three tensile samples were tested for every condition. The angular relationship between the tensile direction and the BD was 0° for all tensile specimens.

2.3. *Simulation model for quantification of melt pool geometries, temperature fields, and solidification parameters*

The FE solver ABAQUS (Dassault Systèmes SE, France) was used in combination with a moving volumetric heat flux user subroutine DFLUX and a gaussian laser intensity profile [30] to numerically simulate transient temperature fields, melt pool geometries, and solidification parameters. The model was discretized using a control volume method in a solution domain that consisted of the powder bed and solidified material. Fig. 2a shows the simulation model with two solution domains and the bidirectional scan strategy of five simulated laser tracks. A simulation of five adjacent laser tracks was chosen to approximate the effect of a laser off-time between two laser tracks. The laser off-time was 50 ms for the M&S condition and 190 µs for all other conditions. The powder bed and solidified material were approximated as homogenous and continuous domains. The temperature dependent heat conductivity was adopted from experimental measurements of a similar X70Mn22 alloy [31]. The liquidus ($T_{liquidus}$ = 1426.23 °C) and solidus ($T_{solidus}$ = 1375.85 °C) temperature, as well as the temperature dependent density, specific heat capacity, and latent heat, were thermodynamically calculated by Thermo-Calc using the TCFE9 database. The powder properties were determined considering the thermo-physical parameters of the solid and the powder bed density, which was defined to be 40% according to Spierings et al. [32]. A variable (φ) for the state change of powder (φ = 0) to solid material (φ = 1) was introduced. By using the user subroutine USDFLD the powder density and the powder heat conductivity were changed linearly after surpassing $T_{liquidus}$ until $T_{solidus}$, until the properties of solidified material are reached.

During L-PBF, the laser radiation interacts mostly with the liquid melt pool [16]. Therefore, an absorption coefficient of 0.41 [33] for liquid iron irradiated by an Nd-YAG laser was chosen. Temperature-

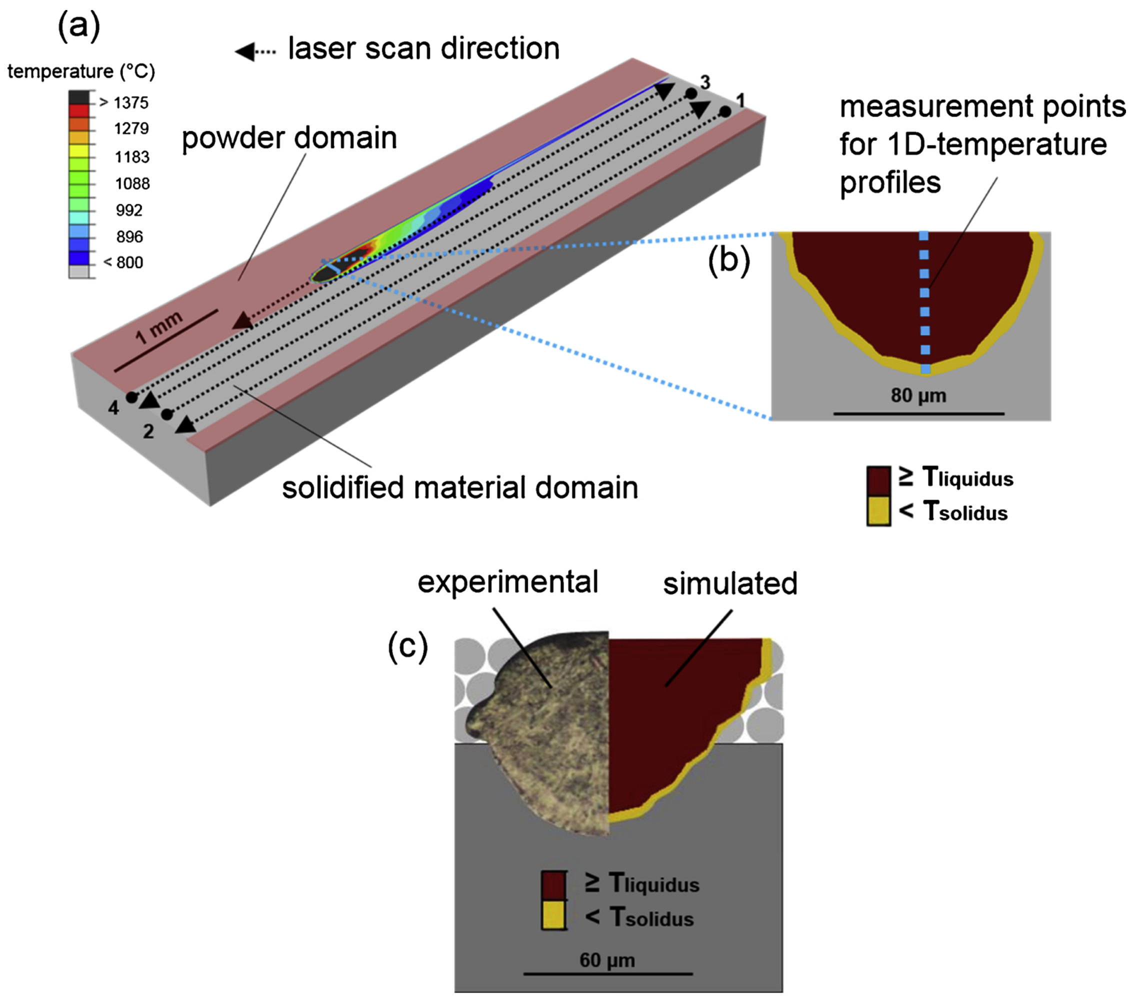

Fig. 2. (a) FE-model for simulation of transient temperature fields, melt pool geometries, and solidification parameters during L-PBF of X30Mn21 HMnS. (b) Cross-section of a simulated melt pool and the position of an 1D-temperature profile for calculation of temperature-time curves and solidification parameters. (c) Validation of the simulation model by comparison of the simulated melt pool size with experimentally generated single tracks for the v950 condition.

time curves, the cooling rate between $T_{liquidus}$ and $T_{solidus,}$ the thermal gradient, and the solidification velocity were calculated in 10 µm steps at the centerline of the cross-section of the melt pool, as schematically shown in Fig. 2b. The cooling rate between $T_{liquidus}$ and $T_{solidus}$ was derived from temperature-time curves. The thermal gradient was calculated considering the distance between $T_{liquidus}$ and $T_{solidus}$ in the melt pool model. The solidification velocity R was calculated considering the inclination of the liquid-solidus interface of the melt pool model and the relationship $R = v \times \cos\alpha$ [34,35], where v is the laser speed and α the angle between the normal of the solid-liquid interface and the direction of the laser beam. The melt pool model was validated by comparison of the simulated melt pool size with experimentally produced single tracks (Fig. 2c). The simulated melt pool sizes revealed an average deviation of less than 20% compared to L-PBF produced single tracks. To ensure computational efficiency, thermal induced volume changes, radiation losses, and the effect of vaporization on heat loss and composition change were neglected. The effect of convective fluid flow on the heat conductivity within the melt pool was simulated by increasing the heat conductivity in the melt pool following the model by Lampa et al. [36].

3. Results

3.1. Influence of L-PBF process parameters on temperature fields, melt pool geometries, and solidification parameters

Fig. 3a and b reveal significant changes in the simulated temperature fields and melt pool sizes of the alloy processed with different L-PBF parameters. Low melt pool depths, melt pool elongations, and higher inclinations of the melt pool resulted in high cooling rates, thermal gradients and solidification velocities (Fig. 3d-g). The temperature-time curves (Fig. 3c) exhibited a significantly lower heat accumulation until the middle of the fifth laser track for M&S condition, as compared to the other states, especially for the ones that were preheated. Cooling rates (Fig. 3d), temperature gradients (Fig. 3e) and the G/R ratios (Fig. 3g) were the highest at the melt pool bottom (fusion line) and the lowest at the upper melt pool surface. In contrast, the solidification velocity (Fig. 3f) was the highest at the upper melt pool surface and extremely low at the fusion line. Increasing the laser speed had a significant influence on the solidification velocity (Fig. 3f), but a minor effect on the thermal gradient (Fig. 3e). Using an off-time of the

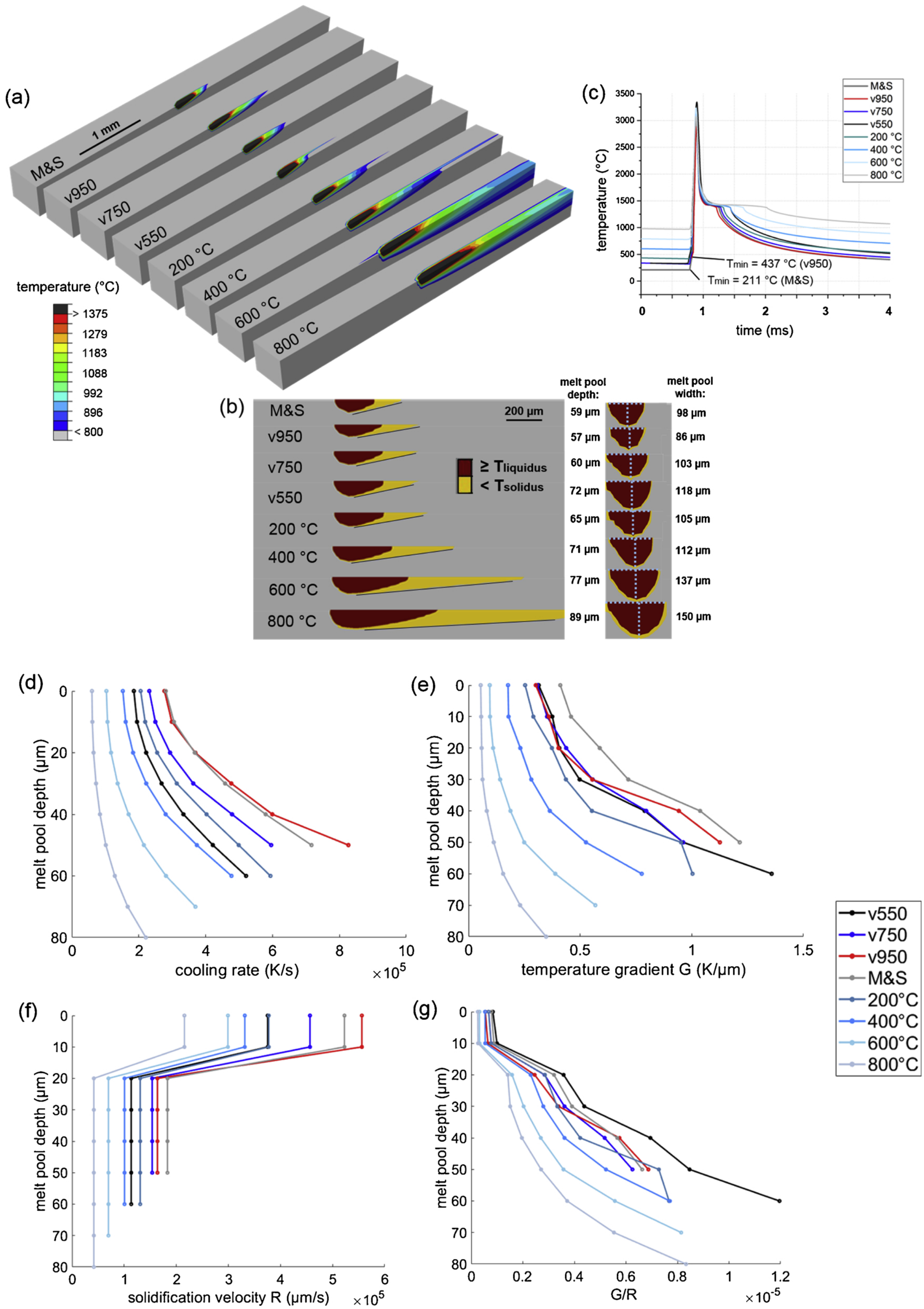

Fig. 3. (a) Isometric sections of simulated temperature fields during L-PBF of X30Mn21 HMnS. (b) Longitudinal section and cross-section of the melt pools. (c) Temperature-time curves of respective measurement points in the center of melt pools. (d–g) Cooling rates, temperature gradients (G), solidification velocities (R) and G/R ratios along the respective centerlines of the melt pool cross-sections. The analysis of melt pool geometries, temperature-time curves, and solidification parameters was performed in the middle of the fifth laser track.

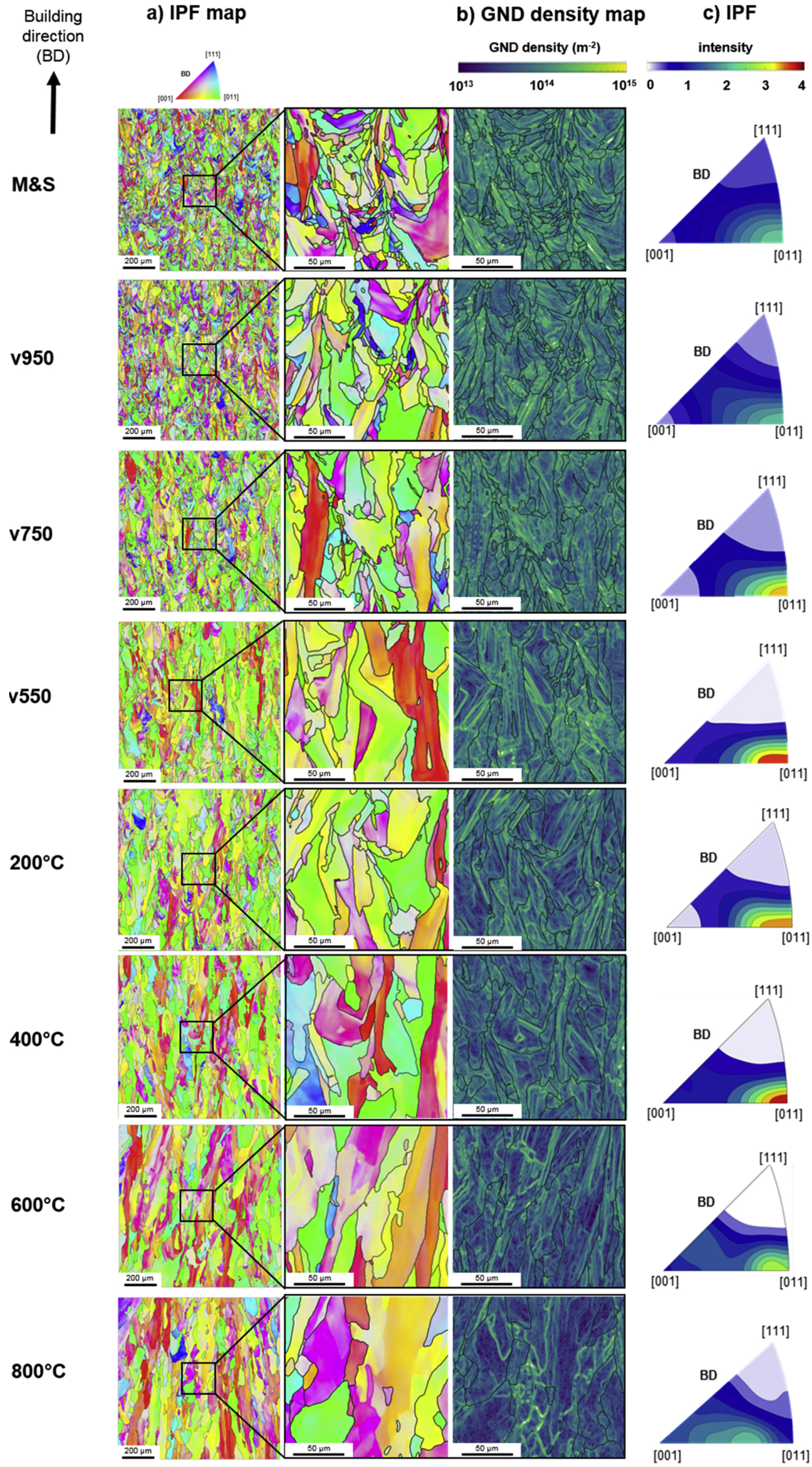

Fig. 4. Grain structure and microtexture of the alloy processed with varying laser speed, preheating temperatures of the build platform and an alternative M&S scan strategy. A corresponding EBSD map with color-coding according to the inverse pole figure (IPF), GND density map and the IPF plot parallel to the building direction is given for each condition.

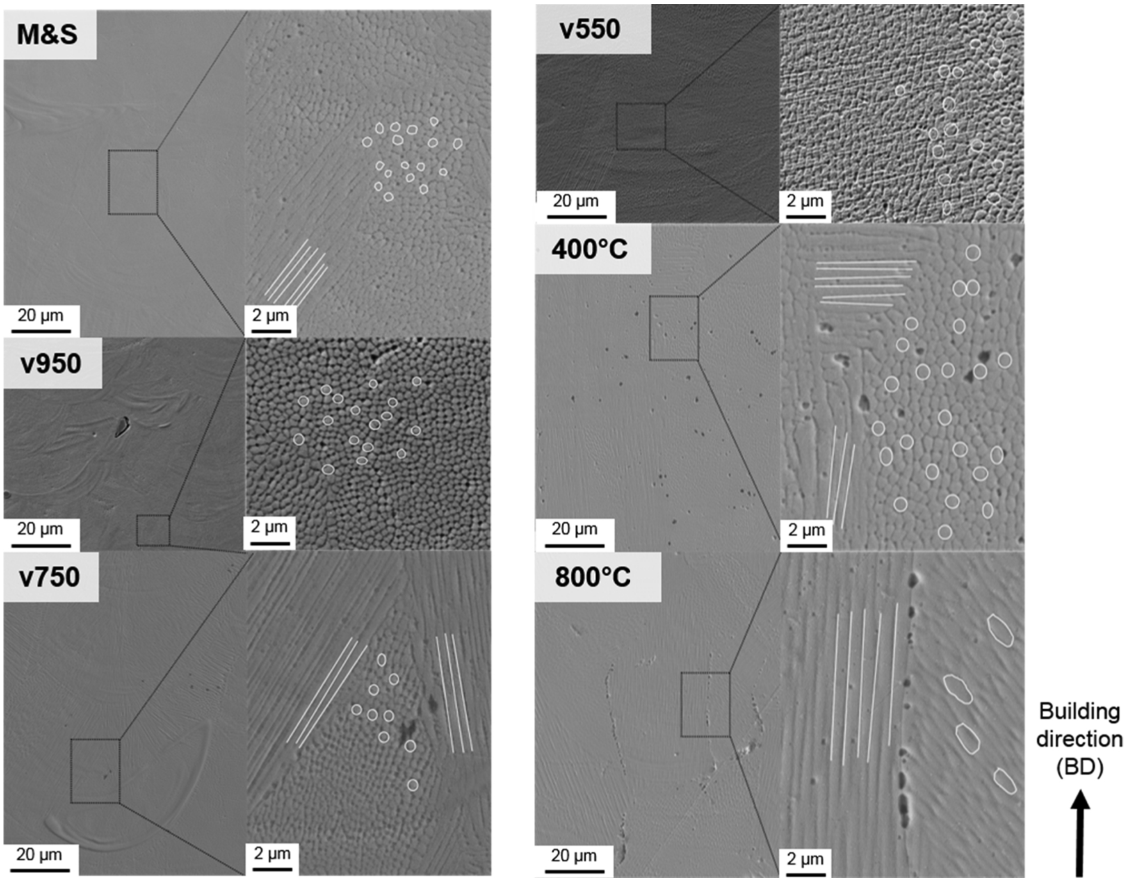

Fig. 5. SE micrographs of L-PBF produced X30Mn21 HMnS in the designated conditions. The white lines and circles reveal cellular solidification structures orientated parallel to the image plane and perpendicular to the image plane, respectively.

laser between five adjacent laser tracks (M&S strategy) strongly increased the thermal gradient and increased the solidification velocity. Higher preheating temperatures decreased both thermal gradient and solidification velocity.

3.2. Influence of L-PBF process parameters on microstructure evolution

EBSD phase analysis of the L-PBF produced X30Mn21 HMnS revealed fully austenitic microstructures in each condition. However, the

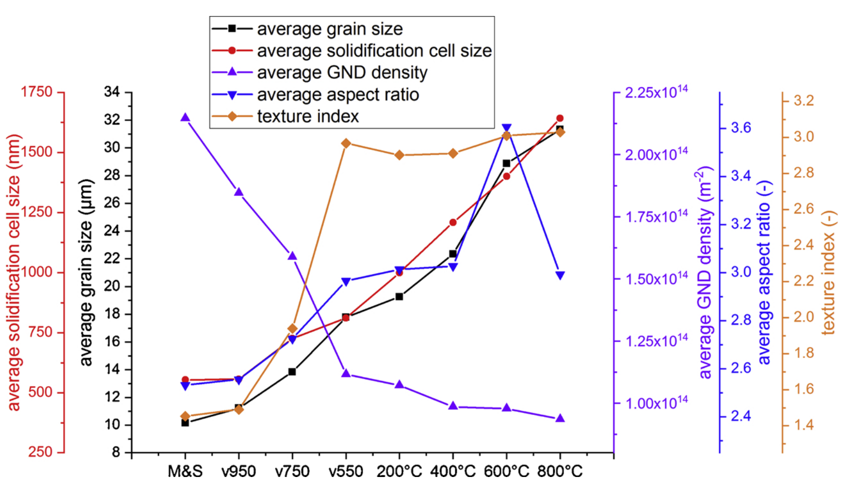

Fig. 6. Dependence of microstructural features on the L-PBF process conditions.

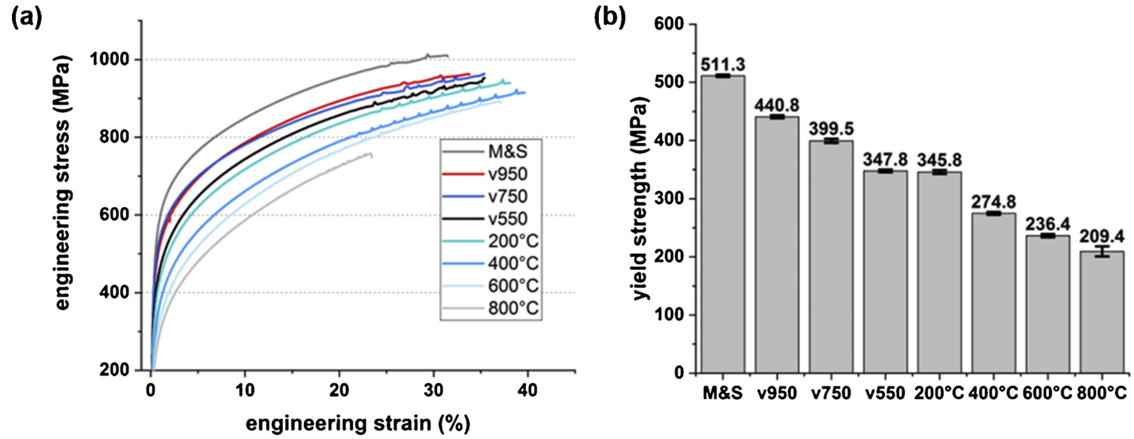

Fig. 7. (a) Engineering stress - engineering strain curves (b) corresponding average yield strengths (Rp0.2). The angular relationship between the tensile direction and the building direction was 0° for all tensile specimens tested.

variation of L-PBF process parameters revealed significant influence on the grain size, grain morphology, texture, and GND distribution (Fig. 4) as well as on the solidification structure (Fig. 5). Quantified values of the microstructural features are given in in Fig. 6 and the appendix (Table A1 and A2).

Firstly, the influence of the laser speed is described. Increasing the laser speed from 550 mm/s to 950 mm/s resulted in a decrease of the average grain size from 17.78 µm (v550) to 11.23 µm (v950). The elongated shape of the grains parallel to the BD remained for all laser speeds, but the extent of elongation decreased at higher laser speeds from an average grain aspect ratio of 2.97 (v550) to 2.56 (v950). The IPF maps and IPF plots given in Fig. 4 indicate the crystal orientations of the grains with respect to the BD. For the lowest laser speed (v550), a large number of grains was oriented with < 101 > parallel to the BD, resulting in a pronounced < 101 > ‖ BD texture with a texture index of 2.97. With increasing laser speed (v950), the < 101 > ‖ BD texture diminishes and becomes more random with a texture index of 1.49. The variation of L-PBF process parameters revealed a significant influence on the GND density (Fig. 4b). GNDs were not distributed homogeneously within the grains, but facilitated the formation of substructures, as indicated by local regions of high GND density (bright regions in Fig. 4). Increasing the laser speed increased the average GND density by 65% from $1.12 \times 10^{14}\,m^{-2}$ to $1.85 \times 10^{14}\,m^{-2}$. The SE micrographs in Fig. 5 reveal the influence of the laser speed on the solidification structure. The approximated positions of the fusion lines are marked in blue. At higher magnification, a solidification cell structure within the melted tracks became visible. Grains consisted of colonies of solidification cells with the same orientation. The grains were elongated parallel to the BD and often extended across multiple laser-melted layers. Solidification cell structures that were orientated mostly perpendicular to the image plane of the SE micrograph were circular or oval shaped (highlighted by white circles). Longitudinal solidification cell structures, which align mostly parallel to the image plane, are highlighted by white lines. The average solidification cell size was decreased from 812 nm (v550) to 557 nm (v950).

Secondly, the influence of the scan strategy is described. The M&S strategy affected the microstructure in a similar way as observed when increasing the laser speed. A reduction of the average grain size and average grain aspect ratio from 13.84 µm to 10.16 µm and from 2.73 to 2.53, respectively, was observed, as compared to the scan strategy without off-time of the laser between two adjacent scan tracks (v750). Furthermore, a weaker random texture (texture index 1.45) compared to the more pronounced < 101 > ‖ BD texture of the v750 sample with a texture index of 1.94 was observed. The GND density was increased by 35% to a value of 2.15×10^{14} and the solidification cell size decreased by 24% (554 nm).

Thirdly, preheating the build platform exhibited contrary effects on the microstructure to increasing the laser speed and varying the scan strategy. With an increase in build platform temperature, grains became coarser and more elongated. The average grain size was increased by 126% and average grain aspect ratio by 10% compared to v750, when using a preheating temperature of 800 °C. Between 200 °C and 800 °C, the texture index remained comparatively constant in the range between 2.90 and 3.03. Until a preheating temperature of 400 °C a < 101 > ‖ BD texture was still visible but decreased in intensity. For a preheating temperature of 800 °C, a texture between < 100 > ‖ BD and < 101 > ‖ BD was observable. Preheating of the build platform exhibited the strongest effect on GND density, which decreased by 41% to a value of $9.36 \times 10^{13}\,m^{-2}$. Also, the solidification cell sizes increased by 126% to a value of 1.6 µm. For a preheating temperature above 200 °C, the formation of randomly distributed pores was observed. Higher preheating of the build platform (800 °C) featured an increased number of pores that were preferentially accumulated at grain boundaries (Fig. 5).

3.3. Mechanical properties

The mechanical properties of the investigated samples are summarized in Fig. 7. Although the yield strength (Rp0.2) was significantly influenced, the flow behavior (Fig. 7a) was similar for all specimens processed with varying L-PBF parameters, i.e. the typical linear work-hardening of HMnS was observed. Increasing the laser speed from 550 mm/s to 950 mm/s resulted in an increase of the yield strength by 26.9% from 347 MPa to 440 MPa, whereas the total elongation decreased by 13.4%. The average ultimate tensile strength was similar, between 951.3 MPa and 952.9 MPa. The specimens produced with the alternative M&S scan strategy revealed the highest yield and tensile strength. In contrast, preheating of the build platform revealed an inverse effect on the mechanical properties. With increasing preheating temperature, the yield and tensile strength decreased, as compared to

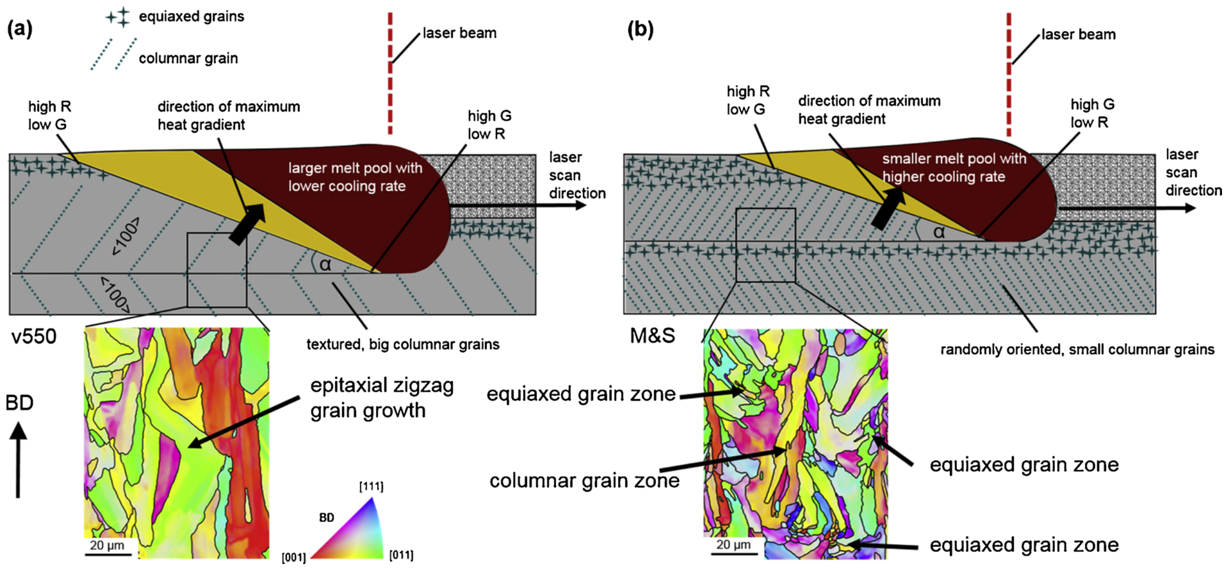

Fig. 8. EBSD IPF maps in combination with schematic longitudinal cross-sections of the melt pool for the (a) v550 condition with a strongly textured elongated zigzag grain structure and (b) the M&S condition with alternated equiaxed and columnar grain zones.

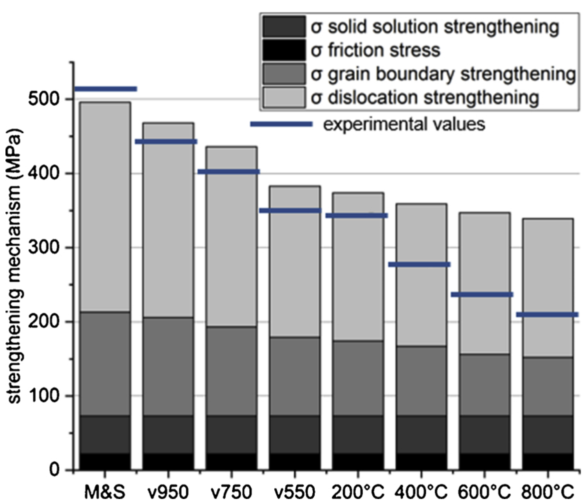

Fig. 9. Comparison of calculated and experimentally determined yield strengths of L-PBF produced HMnS.

the specimens produced without preheating (v750).

4. Discussion

4.1. Microstructure evolution

The focus of the present study was to gain a deeper understanding of the correlation between L-PBF process parameters, microstructure evolution and mechanical properties. The microstructural evolution of L-PBF produced metals is based on the melting of a powder bed by a moving laser source and the solidification of a characteristic elongated melt pool by dissipating heat to the build platform. The temperature field near the solid-liquid interface, in addition to the chemical composition, determine the morphology and size of grains, the solidification structures within the grains, dislocation density, and the crystallographic texture of each L-PBF produced material track. The temperature field directly determines the shape of the melt pool and the corresponding local solidification parameters, i.e. temperature gradient, solidification rate and cooling rate [33,36,37]. In this study the solidification parameters were experimentally governed by variation of laser speed, build platform preheating temperature, and the off-time between two adjacent laser tracks and were quantified by FEM simulations using a thermal solver.

High laser speeds and reduced build platform preheating temperatures are related to shallow and thin melt pools as well as high cooling rates. In contrast, low laser speed and high preheating are related to deeper and wider melt pools and lower cooling rates [37,38]. The simulated melt pool shapes and cooling rates for the applied L-PBF process parameters are consistent with these basic assumptions (Fig. 3b).

During L-PBF the grain morphology and crystallographic texture are determined by a characteristic epitaxial grain growth at the solid-liquid interface of the melt pool. Grain growth occurs preferentially in specific material-dependent crystallographic directions along the maximum heat gradient [35,38,39], as schematically shown in Fig. 8. During L-PBF of the X30Mn21 steel, every deposited layer exhibited the same chemical composition as the previous layer and the material remained fully austenitic from solidification until cooling to room temperature. Under these circumstances, the grains grow epitaxially from the previously deposited layer [36,40], which is partially re-melted. An increased melt pool size facilitates deeper penetration of the melt pool into previous layers. These conditions contribute to pronounced epitaxial growth and thus, to an increased morphological and crystallographic texture [41]. This behavior was also observed in the current study for L-PBF parameters with larger melt pool sizes (v550, 400 °C, 600 °C, 800 °C, Fig. 3). Depending on the anisotropy of the solid-liquid interfacial free energy and interfacial stiffness for different crystal planes, one of the six $< 100 >$ directions of the fcc HMnS is preferred during competitive grain growth [37,42]. Therefore, grains with $< 100 >$ orientation parallel to the maximum temperature gradient in the previously deposited layer serve as a nucleus for epitaxial growth in the newly deposited layer, where the preferred $< 100 >$ orientation is retained due to a growth advantage. Other grain orientations grow slower, which results in the typical columnar and < 100 > -oriented grains of L-PBF produced fcc metals [43].

However, the HMnS steel in this work exhibited a < 110 > fiber texture parallel to the BD. < 110 > fiber textures in the as-built condition were observed in previous studies of additively manufactured fcc metals, e.g. [9,20,44]. This deviation from the expected < 100 > fiber texture is governed by the inclination of the solid-liquid interface and a resulting maximum temperature gradient that is not parallel but inclined by about 25° to the BD [45]. Fig. 8a reveals a schematic longitudinal cross-section of the melt pool in L-PBF, the solid-liquid interface and microstructure of the v550 condition. The inclination of the solid-liquid interface due to the moving laser source and epitaxial grain growth direction are visualized. Depending on the laser scan directions of alternating layers, a zigzag pattern of elongated grains can be observed, caused by alternating maximum thermal gradient directions [38]. Despite the 33° rotation between consecutive layers, epitaxial grain growth and zigzag pattern were observed in the v550 condition, which led to a strong < 110 > fiber texture and a high texture index. The grain size, the < 110 > fiber texture and the texture index for the high cooling rate conditions v950, v750 and M&S were strongly reduced. In order to understand this change in microstructural and textural features, a schematic longitudinal cross-section of the melt pool and the corresponding microstructure of the M&S condition is shown in Fig. 8b. Governed by the smaller melt pool size, the fine-grained microstructure with almost random texture was formed due to the presence of equiaxed grains in the upper part of the deposited tracks. The high cooling rates determine the grain size in the equiaxed zone, which serves as a fine-grained, almost randomly textured substrate for epitaxial growth in the following deposited layer.

The occurrence of an equiaxed grain zone can be understood by consideration of the stability of the solidification front. The stability of the solidification front is governed by the extent of constitutional supercooling, which depends on the material composition (a wider freezing range promotes the formation of an equiaxed grain zone [37]) and the solidification parameters. Lowering the ratio between the thermal gradient and the solidification velocity (G/R), and therefore increasing the constitutional supercooling, changes the solidification morphology from planar to columnar dendritic to cellular and equiaxed dendritic [35]. Due to the complex nature of heat flow in the melt pool, the spatial distribution of temperature is not linear. Therefore, the solidification parameters and the G/R ratio typically vary along with the solid-liquid interface of the melt pool [34], which was validated by FEM simulations (Fig. 3). The thermal gradient G was the lowest near the upper melt pool surface and the highest at the fusion line due to the faster heat transfer through the already solidified material. The solidification velocity R was determined considering the laser speed v and the angle α between the normal of solid-liquid interface and the laser direction. Taking the shape of the simulated melt pools (Fig. 3) into account and using $\mathbf{R} = \mathbf{v} \times \mathbf{cos}\boldsymbol{\alpha}$ [35,36], R was minimal at the fusion boundary and maximal close to the melt pool surface, which is consistent with other works [35,46]. Therefore, especially at the upper part of the melt pool, the G/R conditions for equiaxed grain growth can be satisfied due to the lower thermal gradient and higher R, as compared to the conditions at the fusion line [46]. The FEM melt pool simulations showed the G/R ratio up to fourteen times lower at the top of the melt pool compared to the bottom.

A partial columnar-to-equiaxed transition (CET) occurred under specific process conditions in the investigated material. In order to attain a small mean grain size due to partial CET, the L-PBF-process parameters must be selected in such a way that (i) equiaxed dendritic solidification is promoted and (ii) complete re-melting of the equiaxed zone is suppressed. These prerequisites were fulfilled in the conditions M&S, v950, and v750 and thus, resulted in a strongly reduced grain size and smaller grain aspect ratios, enabled by epitaxial growth from the equiaxed grains at every layer. An off-time of the laser between adjacent laser tracks reduced the heat accumulation and significantly increased the thermal gradient and solidification velocity. Changing the laser speed had a significant influence on the solidification velocity but a minor impact on the thermal gradient. Both conditions resulted in smaller melt pools and higher cooling rates. It must be considered that only one layer with five laser-melted tracks was simulated, whereas the experimentally overserved microstructures are the result of a significantly higher number of laser-melted tracks and layers. However, the cooling rate of the M&S condition was comparable to the v950 condition already after five simulated laser tracks. The effect of reduced heat accumulation and higher cooling rate due to the applied M&S scan strategy is supposedly more pronounced in a larger build volume, i.e. in the physical specimen, due to a significantly higher number of laser-melted tracks. This assumption is supported by the lower grain size and higher GND density of the M&S condition. On the other hand, during the processing of the v550 condition and samples with build platform preheating, higher melt pool depths and lower cooling rates promoted either complete re-melting of the equiaxed grain zone or suppression of its formation, resulting in larger grain sizes and higher aspect ratios (Figs. 3, 6 and 8b). As of yet, a comparable adjustment of grain structure and texture was only achieved by electron beam melting (EBM) [47].

4.2. Strengthening mechanisms

Variation of the solidification conditions influenced not only the size and morphology of microstructural and textural features but also the mechanical properties, especially the yield strength. The experimental mechanical properties are considered as representable due to the low standard deviation of mechanical properties (Table A4). The yield strength of conventionally produced HMnS with a comparable chemical composition is usually in the range between 250–350 MPa [48,49]. The L-PBF produced HMnS in this study revealed a yield strength between 209 MPa to 511 MPa (Fig. 7) due to the accumulation of strengthening mechanisms. The yield strength of undeformed, precipitation-free alloys is usually described considering the contributions of the individual strengthening mechanisms, i.e. friction stress ($\boldsymbol{\sigma}_{\mathbf{i}}$), solid solution strengthening ($\boldsymbol{\sigma}_{\mathbf{SS}}$) and grain boundary strengthening ($\boldsymbol{\sigma}_{\mathbf{GB}}$).

The friction stress, a grain-size-independent parameter, can be interpreted as stress opposing the passage of dislocations along a slip plane and was assumed to be 22 MPa according to Kusakin et al. [50]. The contribution of solid solution strengthening $\boldsymbol{\sigma}_{\mathbf{SS}}$ was calculated to be 51 MPa using a semi-phenomenological model [48]. The grain boundary strengthening effect $\boldsymbol{\sigma}_{\boldsymbol{GB}}$ due to dislocation pile-up is described by the Hall-Petch relationship $\boldsymbol{\sigma}_{\mathbf{GB}} = \mathbf{k}_{\mathbf{HP}}/\sqrt{\mathbf{d}}$ where $\mathbf{k}_{\mathbf{HP}}$ is the Hall-Petch parameter and $\boldsymbol{d}$ the average grain size. The Hall-Petch parameter was experimentally validated in other studies to be 445

MPa × $\mu m^{1/2}$ [12,13,51–53]. Using the measured grain sizes from the microstructural analysis and $\sigma_{YS} = \sigma_i + \sigma_{SS} + \sigma_{GB}$, the yield strength was strongly underestimated for all L-PBF process parameters applied. Therefore, an additional strengthening mechanism must be active, especially for specimens subjected to processing with a high cooling rate. Previous investigations on LBPF-produced AISI 316 L proposed that the sub-micron solidification cell structure is a significant contributor to the enhanced yield strength, which was correlated with the solidification cell size based on the Hall-Petch relationship [54]. By employing this hypothesis, the estimated yield strength values were much higher than the experimentally determined ones. Therefore, the contribution of the solidification structure cannot be described by the Hall-Petch relationship and is considered negligible, due to low misorientation angles between adjacent cells. However, other studies reported high dislocation densities within solidification cell boundaries [54–56]. In this work, the size of the solidification cells was significantly reduced with higher cooling rates, which was consistent with the work of Kundin et al. [21] and coincided with the development of high GND densities (Fig. 6). These dislocations can be considered as effective obstacles to hinder the movement of mobile dislocations and thus, are considered as contributing factor to the strength.

The contribution of dislocation strengthening (σ_ρ) and σ_{GB} to the yield strength is described by a summation of the individual strengthening mechanisms as follows: $\sigma_{YS} = \sigma_i + \sigma_{SS} + \sigma_{GB} + \sigma_\rho$ [57,58]. To calculate the contribution of the experimentally measured GND densities on the yield strength the Taylor equation $\sigma_\rho = \alpha \times \mathbf{M} \times \mathbf{G} \times \mathbf{b} \times \sqrt{\rho}$ was applied [59]. A dislocation interaction factor α of 0.3 was chosen. The shear modulus $\boldsymbol{G}$ of 68.34 GPa was calculated by using the linearly regressed model suggested by Lee et al. [59]. The corresponding Taylor factors $\boldsymbol{M}$ were calculated from EBSD data using the MATLAB toolbox MTEX. A burgers vector $\mathbf{b}$ of $2.5503*10^{-10}$ m was calculated considering lattice distortion by alloying elements and temperature [60]. The total dislocation density ρ is comprised of both geometrically necessary dislocations (GNDs) and statistically stored dislocations (SSDs) [61]. GNDs, are immobile dislocations, which carry and store lattice curvature resulting from local deformation gradients [61–65], which are caused by thermally induced volume changes during solidification of the small melt pool during L-PBF. L-PBF parameters resulting in high cooling rates and high thermal gradients revealed an increased formation of GNDs (Fig. 6). As suggested by Kubin [65] and Smith et al. [66] GNDs account for most of the total dislocation population in plastically deformed fcc metals. The cold, fixed build platform and pre-deposited layers constrain the thermal expansion and compression of the newly deposited material during the cyclic heating and cooling during L-PBF. This mechanical constraint is accommodated by both elastic and plastic strains as the newly deposited HMnS cools down [67]. Higher cooling rates and especially stronger thermal gradients increased this effect causing a higher GND density. In contrast, the lower cooling rates and thermal gradients of the conditions with higher preheating temperatures of the build platform led to lower GND densities after solidification. It can be assumed that the GND density of already solidified layers was additionally reduced due to recovery processes. Furthermore, grain growth in the solidified layers subjected to preheating can be neglected, since this effect is usually observed at higher temperatures in HMnS, e.g. at 1150 °C in [68]. The strengthening effect of high GND densities is controlled by short-range interaction between moving dislocations and immobile GND and long-range stress fields, which are associated with microstructural features, such as solidification cell structures [61].

The calculations of the yield strengths by the summation of the individual strengthening mechanisms exhibited a good agreement with experimental values for high and intermediate cooling rates, i.e. for the conditions M&S, v950, v750, v750 and 200 °C (Fig. 9). For the lower cooling rate L-PBF conditions (400 °C, 600 °C, 800 °C) the yield strengths were overestimated. Following [67], we assume that the lower bound for a quantitative acquisition of GND densities by EBSD was reached for the used EBSD settings and specimens with inherently lower GND densities. Therefore, the overestimation of the GND densities for the conditions processed with lower cooling rates resulted from an overestimation of dislocation strengthening.

Haase et al. [20] reported an anisotropic deformation behavior of L-PBF produced X30Mn21 HMnS. The yield and tensile strength increased, whereas the total elongation decreased with an increasing angle between the building direction and tensile direction. The experimental characterization of anisotropy of the mechanical properties was not in the scope of the present study. However, since the morphological and crystallographic texture of the conditions M&S and v950 were reduced, a reduction of anisotropy can be expected. On the other hand, the anisotropy is presumably increased for conditions with strong texture and elongated grains, i.e. v550, 600 °C, 800 °C.

Due to a better understanding of the effect of L-PBF process parameters on the microstructural evolution, the properties of L-PBF produced HMnS can be tailored for different applications. The promotion of low grain size and weak texture due to the formation of equiaxed grain zones can be used for higher yield strength, a more isotropic deformation behavior and increased resistance against crack propagation. The adjustment of textured columnar grains can improve the creep resistance of parts applied at high temperature.

5. Conclusions

In this work, the variation of L-PBF process parameters, i.e. laser speed, build platform preheating temperature and an alternative M&S scan strategy, was used to manipulate the microstructure, texture and the yield strength of an austenitic AHSS. The combination of experimental and computational methods facilitated a better fundamental understanding of the correlation between process, microstructure and properties. The following conclusions can be drawn:

I Melt pool simulations allowed for precise quantification of the melt pool geometry and solidification parameters. An off-time of the laser between adjacent laser tracks (M&S scan strategy) reduced the heat accumulation and significantly increased the thermal gradient resulting in a smaller melt pool and higher cooling rates. Preheating of the build platform decreased the thermal gradient and promoted deep and elongated melt pools with low solidification velocities and reduced cooling rates. Changing the laser speed had a significant influence on the solidification velocity but a minor impact on the thermal gradient.

II A low penetration depth of the melt pool, a low ratio between the thermal gradient and the solidification velocity, and a high cooling rate were favorable for microstructure refinement and reduced texture intensity, due to a partial change in the solidification mode from columnar to equiaxed.

III Different strengthening mechanisms contributed to the yield strength of all conditions. Dislocation strengthening due to high geometrically necessary dislocation (GND) densities in the as-built state revealed the strongest contribution to the yield strength, followed by grain boundary strengthening.
IV High thermal gradients and cooling rates resulted in high GND densities, lower grain sizes, solidification cell sizes, grain aspect ratios and texture indices, and vice versa. These correlations can be effectively used to tailor the mechanical properties of additively manufactured AHSS.

Declaration of interest

None.

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Appendix

Table A1
Quantitative analysis of microstructural features of the investigated L-PBF produced steel.

	M&S	v950	v750	v550	200 °C	400 °C	600 °C	800 °C
average grain size (μm)	10.16 ± 10.23	11.23 ± 11.78	13.84 ± 15.37	17.78 ± 23.46	19.26 ± 23.58	22.34 ± 28.90	28.86 ± 36.72	31.34 ± 39.55
average grain aspect ratio (-)	2.53 ± 1.29	2.56 ± 1.36	2.73 ± 1.54	2.97 ± 1.69	3.01 ± 1.76	3.03 ± 1.82	3.61 ± 2.05	2.99 ± 1.68
average solidification cell size (nm)	554 ± 167	557 ± 145	726 ± 260	812 ± 301	987 ± 387	1209 ± 571	1412 ± 612	1642 ± 684
GND density (m^{-2})	2.15E + 14	1.85E + 14	1.59E + 14	1.12E + 14	1.07E + 14	9.86E + 13	9.78E + 13	9.36E + 13
texture index (-)	1.45	1.49	1.94	2.97	2.90	2.91	3.01	3.03
number of grains per mm^2 (-)	4976	3921	2518	1244	1227	820	515	450

Table A2
Selected changes of average grain size, grain aspect ratio, solidification cell size, GND density, and texture index.

Change in	scan speed v550 to v950	scan strategy v750 to M&S	build platform heating v750 to 800 °C
average grain size	− 37%	− 27%	+ 127%
average grain aspect ratio	− 14%	− 7%	+ 10%
average solidification cell size	− 31%	− 24%	+ 126%
average GND density	+ 65%	+ 35%	− 41%
average yield strength	+ 27%	+ 28%	− 91%
texture index	− 31%	− 24%	+ 56%

Table A3
Relative density measurements of all conditions obtained by optical measurement of the residual porosity.

	M&S	v950	v750	v550	200 °C	400 °C	600 °C	800 °C
density (-)	99.98	99.93	99.98	99.98	99.99	99.99	99.99	99.99

Table A4
Quantitative analysis of mechanical properties of the investigated L-PBF produced steel.

	M&S	v950	v750	v550	200 °C	400 °C	600 °C	800 °C
average yield strength (MPa)	511.31 ± 1.99	440.82 ± 2.61	399.50 ± 3.53	347.77 ± 2.42	345.80 ± 3.25	274.79 ± 2.44	236.43 ± 2.85	209.35 ± 8.69
average tensile strength (MPa)	1017 ± 4.98	959.90 ± 17.47	951.27 ± 16.44	952.94 ± 2.48	935.70 ± 8.86	915.40 ± 11.93	873.62 ± 41.33	737.66 ± 30.64
average total elongation (MPa)	31.62 ± 0.66	31.50 ± 4.03	34.47 ± 2.28	36.38 ± 2.38	35.24 ± 1.70	37.98 ± 1.77	33.45 ± 5.79	20.93 ± 3.40

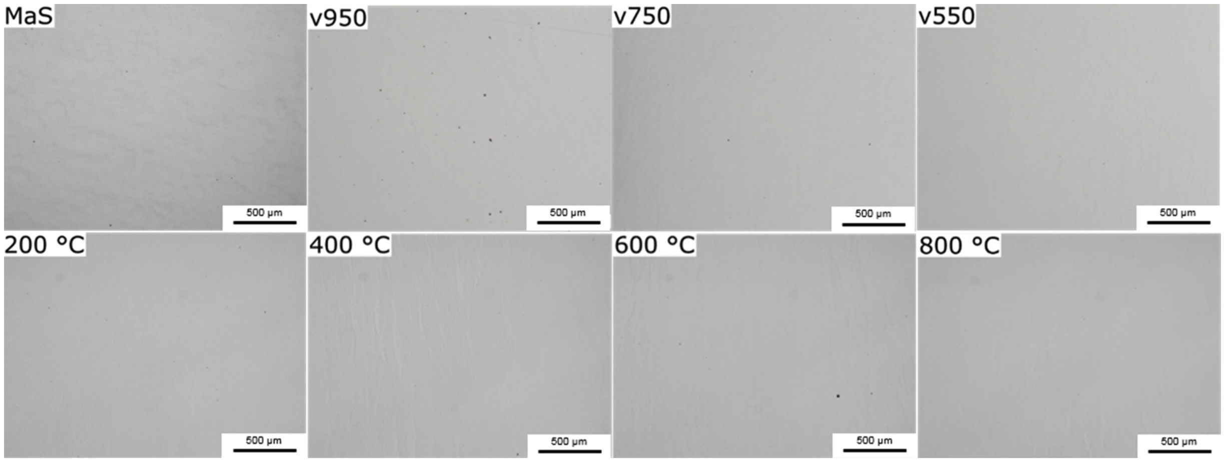

Fig. A1. Optical micrographs of the middle of the cross-section parallel to the building direction for optical measurement of residual porosity.

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